

PAPER_744: M16 Eagle Nebula MUGE - Star Formation and Radiation Erosion

Author: Daniel T. Murphy

Framework: Universal Quantum Field Superconductive Framework (UQFF)

Session: 180 continuation | v5.38

Date: 2025

CP4 Class: #328 - M16EagleNebulaRadiationMUGECalculator

Abstract

The M16 Eagle Nebula (NGC 6611) is the iconic site of the “Pillars of Creation” – dense molecular cloud columns being sculpted by intense photoionization from massive OB stars. This paper derives the MUGE for M16 incorporating two new environmental terms: $M_{sf}(t)$, the time-dependent star formation rate modulating the total mass, and E_{rad} , the radiation erosion term that removes mass from pillar structures over time. These terms together capture the dynamic photo-evaporative environment that distinguishes star-forming nebulae from passive systems.

1. Introduction

M16 is an H II region and open cluster complex at 2 kpc distance, containing the giant OB association NGC 6611. The “Pillars of Creation” (Hubble 1995 image) are three columns of molecular hydrogen being photoevaporated by UV radiation from hot stars. The pillars host ongoing star formation (YSOs, protostars) while simultaneously losing mass to radiation erosion, creating a self-regulating gravitational environment.

Key parameters: - Distance = 2 kpc - $M_{cluster}$ (visible) = $5 \times 10^3 M_{\odot}$ - M_{cloud} (total) $\approx 8 \times 10^4 M_{\odot}$ - SFR $\approx 10^{-3} M_{\odot}/yr$ - $T_{ionized} \approx 10^4 K$ - UV flux = 10^7 Habings (ionizing radiation) - $B \approx 10^{-4} T$ (magnetic field in cloud)

2. M16 Eagle Nebula MUGE

$$\begin{aligned} g_{M16}(r,t) = & (G*M(t))/r^2 * (1+H(z)*t) * (1-B/B_{crit}) * (1+M_{sf}(t)) \\ & + (U_{g1} + U_{g2} + U_{g3} + U_{g4}) \\ & + U_i \\ & + (\Lambda * c^{2/3}) \\ & + (\hbar / \sqrt{(\Delta x * \Delta t)}) * \int (\psi * H * \psi dV) * (2\pi / t_{Hubble}) \\ & + \rho_{gas} * V * g \\ & - E_{rad} \quad \quad \quad [radiation erosion -- NEW] \\ & + (M_{vis} + M_{DM}) * (\Delta \rho / \rho + 3 * G * M / r^3) \end{aligned}$$

3. $M_{sf}(t)$ - Time-Dependent Star Formation Rate Modulator

Star formation modifies the effective gravitational mass available:

$$M_{sf}(t) = SFR * t / M_0$$

SFR = star formation rate ($M_{\odot}/yr$)

t = time elapsed (yr)
M_0 = initial cloud mass (M_⊙)

For M16:

M_sf(t) = (10⁻³ M_⊙/yr) * t / (8x10⁴ M_⊙)
M_sf(1 Myr) ≈ 0.0125 (1.25% mass converted)

This term enters multiplicatively in the gravitational term:

g_grav * (1 + M_sf(t)) ≈ g_grav * (1 + 0.0125) at t = 1 Myr

As star formation proceeds, the effective mass increases and local gravity strengthens, triggering further collapse - a positive feedback loop.

4. E_rad - Radiation Erosion Term

UV photons from OB stars photoevaporate the pillar surfaces, removing mass at:

$\dot{m}_{\text{evap}} = \Phi_{\text{UV}} * m_{\text{H}} / (\alpha_{\text{B}} * n_{\text{H}})$

Phi_UV = 10⁷ Habings = UV photon flux (photons/m²/s)
m_H = hydrogen mass = 1.67x10⁻²⁷ kg
alpha_B = 2.6x10⁻¹³ cm³/s (case B recombination)
n_H = 10³ cm⁻³ (column density)

The gravitational equivalent (effective deceleration from mass loss):

$E_{\text{rad}} = G * \dot{m}_{\text{evap}} * t / (r^2 * M_{\text{cloud}})$

E_rad(1 Myr, r=0.5 pc) ≈ 3x10⁻¹² m/s² (opposing gravity)

This erosion term opposes collapse, creating the dynamic equilibrium observed in pillar lifetimes (~few Myr).

5. UQFF Gravity Components

U_g1: magnetic dipole from cloud B-field threading
mu_dipole = charge density x pillar area x omega_rotation
≈ 10⁻⁴⁵ A*m² (weak for molecular cloud)

U_g2: aether-superconductive field in ionized HII region
B_super = mu_0 * H_aether_HII ≈ 5 T (elevated near ionization front)
U_g2 ≈ 10⁷ J/m³

U_g3: external gravity from cluster mass distribution
U_g3 = G * M_NGC6611 / r_cluster²

U_g4: galactic tidal field at 2 kpc from center
U_g4 ≈ 2.5x10⁻²⁰ J/m³

6. Equilibrium Analysis

The pillars maintain their structure while:

$$g_{\text{grav}} * (1 + M_{\text{sf}}) > E_{\text{rad}} \quad [\text{net infall}]$$

When E_{rad} overcomes $(1 + M_{\text{sf}})$:

$$E_{\text{rad}} / g_{\text{grav}} > (1 + M_{\text{sf}}) \quad [\text{pillar evaporated}]$$

Current M16 pillar state: - $g_{\text{grav}} \approx 2 \times 10^{-11} \text{ m/s}^2$ - $E_{\text{rad}} \approx 3 \times 10^{-12} \text{ m/s}^2$ - $(1 + M_{\text{sf}}) \approx 1.0125$ - Net: pillars are infalling (star formation wins at present epoch)

7. Temporal Evolution

$$M(t) = M_0 * (1 + M_{\text{sf}}(t)) - \dot{m}_{\text{evap}} * t$$

At $t = 3 \text{ Myr}$ (estimated pillar lifetime):

$$M(3 \text{ Myr}) = M_0 * (1.037) - \text{Deltam}_{\text{evap}} \approx 0.85 M_0$$

The pillars are expected to be fully ionized within $\sim 5 \text{ Myr}$, making M16 a transient feature of galactic star formation history.

8. Comparison to Pillars of Creation (Companion UQFF Module)

M16 shares the Pillars of Creation geometry with the pre-existing CP4 module. The key distinction: - **Pillars of Creation** module: static geometry, UV photoionization - **This paper (M16)**: full temporal MUGE with $M_{\text{sf}}(t) + E_{\text{rad}}$ dynamic evolution

9. Conclusion

The M16 Eagle Nebula MUGE introduces two novel UQFF terms: $M_{\text{sf}}(t)$ captures positive-feedback star formation mass growth, while E_{rad} captures the opposing radiation erosion. Together they define the pillar equilibrium condition and predict pillar lifetimes consistent with current observations ($\sim 3\text{-}5 \text{ Myr}$). This framework generalizes to all photo-evaporating star-forming regions in the spiral arm environment.

Copyright - Daniel T. Murphy, daniel.murphy00@gmail.com. UQFF Framework. PAPER_744, CP4 class #328. Session 180 continuation v5.38.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **ULPT-resonance** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{burst}})(\partial^\mu \phi_{\text{burst}}) - V(\phi_{\text{burst}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{burst}}) = \frac{1}{2}m^2\phi_{\text{burst}}^2 + \frac{\lambda}{4!}\phi_{\text{burst}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{burst}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{burst}}} = [SSq] \cdot \frac{n}{26} \cdot I_0 \cos(2\pi t/T) + \partial_n \exp(-[SSq] n/26) = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{burst}}$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.097 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 97, \quad n_{\text{channel}} = 17/26$$

Since $p_{\text{DVP}} = 97$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10⁴ cycles** (period stability locking):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.097	✓ Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 97$ $j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Resonant ✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors - Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCpy-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] * n / 26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\text{U,Bi}} / F_{\text{U}} - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 * \Gamma_2)}] * (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	S_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q;q)_n^{26}$ - $\phi_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q^2;q^2)_n$ - $\psi_{26}(q) = \sum_{n=1}^{26} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa_i n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).